INTER IIT TECH MEET 15.0

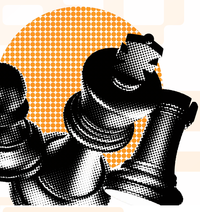
IIT Guwahati Tech Board

Inter IIT Bootcamp

#Gold forGuwahati

Phase 1

Quant PS



- Team Size up to 1-3 Members
- Mentorship Session
- Submission Deadline 7th October

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Inter IIT Bootcamp Quant '26: BTC/USDT and ETH/USDT

1. Domain Description

Algorithmic trading is a domain within the finance industry that uses computer programs and mathematical models to execute trading strategies. It requires a strong understanding of financial markets, data analysis, statistical modeling, and programming.
Participants are expected to apply this expertise to build strategies that outperform benchmarks, protect capital, and remain statistically sound.

2. Problem Description

Participants must develop algorithmic trading strategies for the BTC/USDT and ETH/USDT cryptocurrency markets that outperform benchmark returns while managing risk effectively. Two strategies are required: one for BTC/USDT and one for ETH/USDT. The work is divided into the following tasks.

1. Data Acquisition. Use historical price and trading volume data for BTC/USDT and ETH/USDT from January 1, 2021 to December 31, 2025 for strategy development and testing.
2. Data Preprocessing. Clean and preprocess the data for both trading pairs over the same period. Ensure data quality and consistency, handle missing data, and apply any necessary data transformations.
3. Strategy Design. Design strategies based on machine learning, statistical or mathematical models tailored to the BTC/USDT and ETH/USDT markets.
4. Backtesting. Implement the strategies and backtest them extensively on the historical data of both markets. Simulations must include transaction costs and slippage at a combined rate of 0.15% per transaction.
5. Risk Management. Develop risk management rules and mechanisms specific to each market to protect capital and reduce drawdowns.
6. Optimization. Fine-tune each strategy to maximize returns while keeping risk at acceptable levels, adjusting parameters and rules based on backtesting results.

3. Deliverables

1. Algorithm Code. Source code for the two strategies (one for BTC/USDT, one for ETH/USDT), with documentation explaining the logic and parameters used.
2. Backtesting Results. Detailed backtesting results for each market over the five-year historical period, including performance metrics (e.g., Sharpe ratio, annualized returns, maximum drawdown) and visualizations (e.g., equity curve, trade history).
3. Risk Management Plan. A clear description of the risk management mechanisms used for each market, including stop-loss rules and risk-reward ratios.
4. Report. A comprehensive report on the development process, strategy optimization, and insights gained, covering both BTC/USDT and ETH/USDT.
5. Final Presentation. A presentation, in English, summarizing the strategies for both markets, their rationale, backtesting results, and risk management approach.

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4. Evaluation Criteria

Submissions are scored out of 100 points across six criteria.

No.	Criterion	Points
1	Performance	30
2	Strategy Logic	20
3	Risk Management	15
4	Code Quality and Data Visualization	10
5	Report	15
6	Final Presentation	10
	Total	100

4.1 Performance (30 points)

This criterion evaluates the historical performance of the strategies in both markets, including risk-adjusted returns, consistency, and drawdowns. Excellent strategies are those that:

- maintain narrow drawdowns with good time-to-recovery; and
- beat the benchmark in more than 50% of the quarters

Strategies will also be tested on unseen data, including hidden data segments that are not released to participants. A strategy developed on the training data should perform similarly on the unseen data. Strategies that generalize well will have a distinct advantage in the leaderboard ranking.

4.2 Strategy Logic (20 points)

This criterion assesses the soundness and effectiveness of the mathematical and statistical models used, with a focus on BTC/USDT and ETH/USDT market dynamics.

4.3 Risk Management (15 points)

This criterion evaluates how effective the risk management mechanisms are at preserving capital and maintaining narrow drawdowns in each market.

4.4 Code Quality and Data Visualization (10 points)

This criterion reviews the clarity, organization, and efficiency of the code for both markets and the quality of the data visualizations.
The use of any external backtesting or trading library in the submitted code (for example, backtrader or backtesting.py) is strictly prohibited and will lead to serious penalties in the leaderboard ranking.

4.5 Report (15 points)

This criterion evaluates the completeness and depth of the final report, with emphasis on insights and analysis specific to BTC/USDT and ETH/USDT.

4.6 Final Presentation (10 points)

This criterion assesses the quality and clarity of the presentation materials, with a focus on the BTC/USDT and ETH/USDT strategies.

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5. Recommended Approaches

1. Machine Learning & Predictive Modeling
2. Statistical & Econometric Methods
3. Mean Reverting or Trend-Following Strategies
4. Hybrid / Integrated Frameworks
5. Market Microstructure & Quantitative Dynamics

In the pursuit of implementing profitable ML models for algorithmic trading, use approaches like state-of-the-art explainability techniques, including SHAP, model-agnostic methods such as LIME, Partial Dependence Plots (PDP) etc.

6. Avoiding Overfitting

Participants must guard against overfitting. Overfit strategies can look highly profitable in backtests but lack robustness, and often lose significantly on unseen data and in live trading (forward testing).

7. Output Metrics

The following metrics must be reported:

1. Gross Profit
2. Net Profit
3. Total Closed Trades
4. Win Rate (Profitability %)
5. Max Drawdown
6. Gross Loss
7. Average Winning Trade (USDT)
8. Average Losing Trade (USDT)
9. Buy-and-Hold Return of BTC
10. Largest Losing Trade (USDT)
11. Largest Winning Trade (USDT)
12. Sharpe Ratio
13. Sortino Ratio
14. Average Holding Duration per Trade
15. Maximum Holding Duration per Trade

8. Resources

Basic Resources : https://docs.google.com/document/d/1U6gyxT3jOkVU3v0Ti1uGoPYuMDrDKK-JtyA4daFuE0o/edit?usp=sharing

Advanced: https://docs.google.com/document/d/1ctfMRZQGGoHB1_3K_ey7H19BNL9WxhUzsllaVnMO-c8/edit?tab=t.0 (Ignore SDK)